

✓ Lab 11: Visualizing Time Series Data

```
import pandas as pd
import matplotlib.pyplot as plt
import plotly.express as px
df = pd.read_csv('airline-passengers.csv')
df['Month'] = pd.to_datetime(df['Month'])
df.head()
```



	Month	Passengers
0	1949-01-01	112
1	1949-02-01	118
2	1949-03-01	132
3	1949-04-01	129
4	1949-05-01	121

✓ 1. Line Plot

Description:

A simple line plot showing the number of airline passengers over time.

Features used:

- Month (x-axis, time)
- Passengers (y-axis, value)

Observation:

- There is a clear upward trend and visible seasonality in the number of passengers.

```
plt.figure(figsize=(12,6))
plt.plot(df['Month'], df['Passengers'], marker='o')
plt.title('Monthly International Airline Passengers (1949-1960)')
plt.xlabel('Month')
plt.ylabel('Number of Passengers')
plt.grid(True)
plt.show()
```



✓ 2. Rolling Average Plot

Description:

A rolling (moving) average smooths out short-term fluctuations and highlights longer-term trends.

Features used:

- Month (x-axis)
- Passengers (y-axis)
- 12-month rolling mean

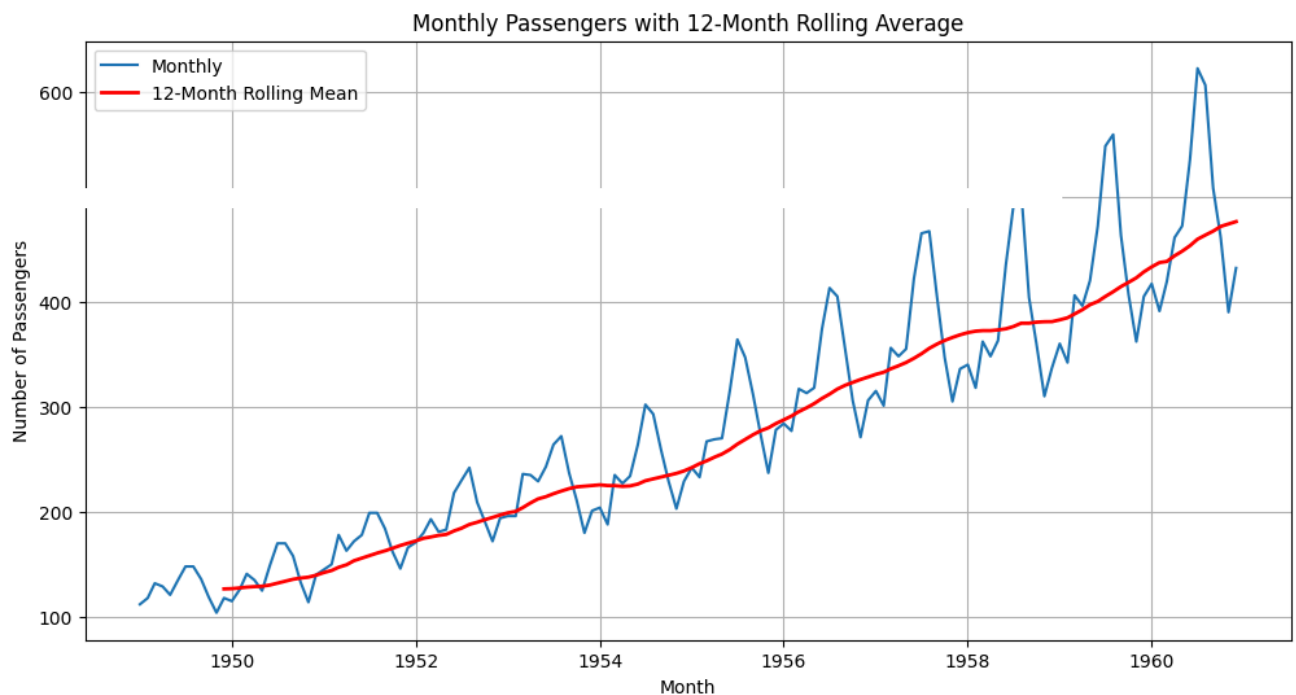
Observation:

- The rolling average line makes the long-term growth trend even clearer.

```
df['RollingMean'] = df['Passengers'].rolling(window=12).mean()
```

```
plt.figure(figsize=(12,6))  
plt.plot(df['Month'], df['Passengers'], label='Monthly')  
plt.plot(df['Month'], df['RollingMean'], color='red', linewidth=2, label='12-Mont
```

```
plt.title('Monthly Passengers with 12-Month Rolling Average')  
plt.xlabel('Month')  
plt.ylabel('Number of Passengers')  
plt.legend()  
plt.grid(True)  
plt.show()
```



✓ 3. Seasonal Decomposition

Description:

Decomposes the time series into trend, seasonality, and residuals using statsmodels.

Features used:

- Month (time)
- Passengers (value)

Observation:

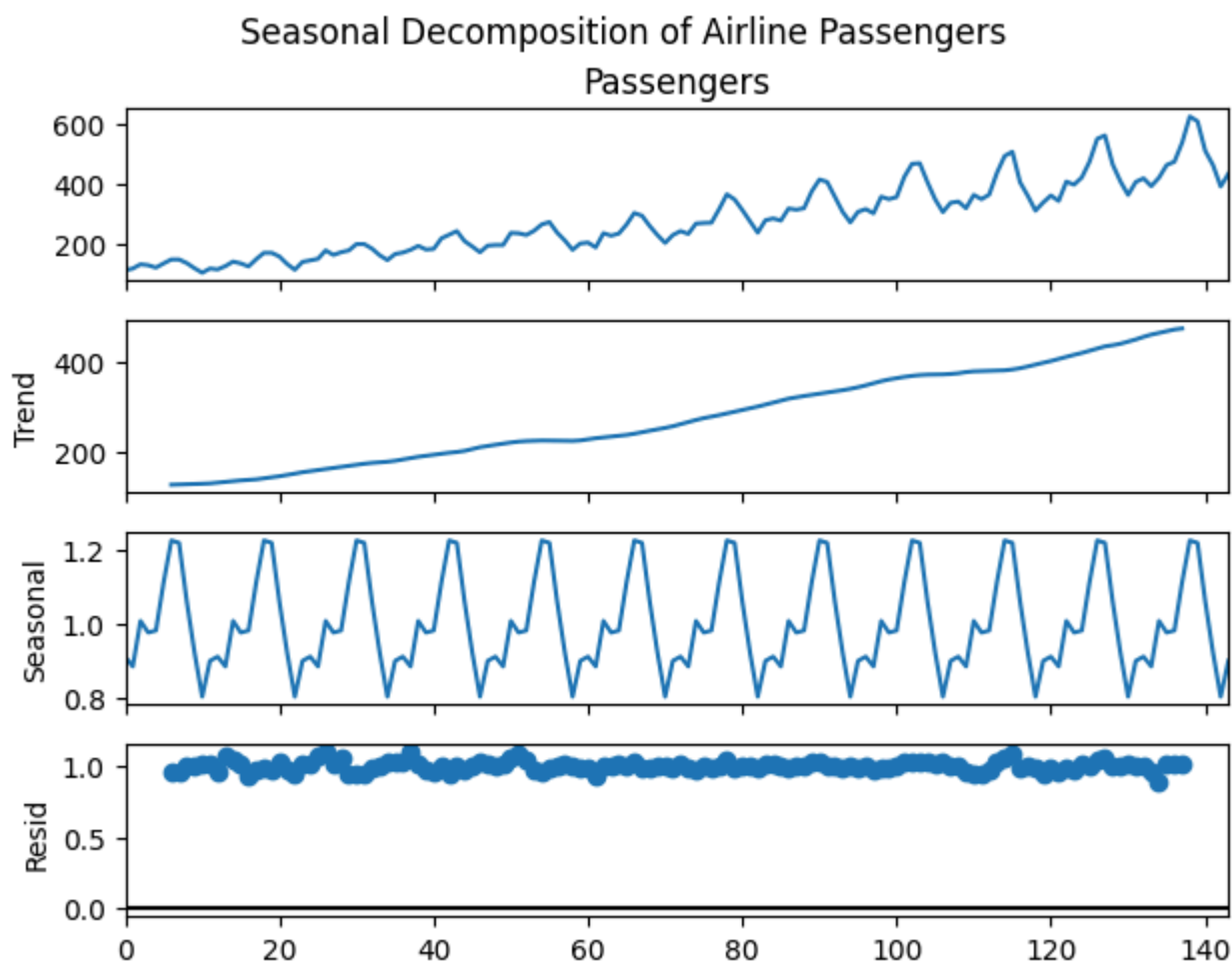
- The decomposition clearly separates the upward trend and repeating seasonal pattern.

```
from statsmodels.tsa.seasonal import seasonal_decompose
```

```

result = seasonal_decompose(df['Passengers'], model='multiplicative', period=12)
result.plot()
plt.suptitle('Seasonal Decomposition of Airline Passengers', y=1.02)
plt.show()

```



✓ 4. Heatmap of Monthly Patterns

Description:

A heatmap showing the number of passengers for each month and year, revealing seasonal patterns.

Features used:

- Year (y-axis)
- Month (x-axis)
- Passengers (color)

Observation:

- The heatmap highlights peak travel months and year-over-year growth.

```
import seaborn as sns
```

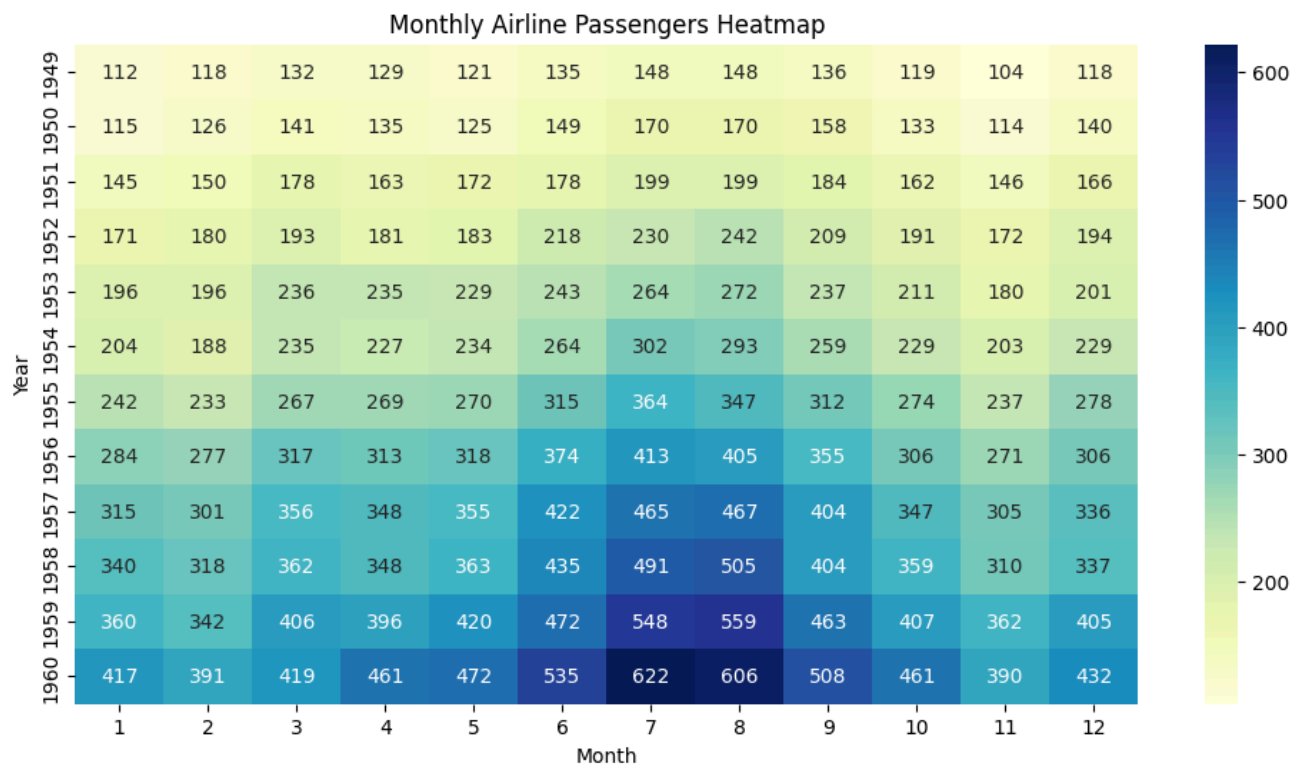
```

df['Year'] = df['Month'].dt.year
df['MonthNum'] = df['Month'].dt.month

```

```
pivot = df.pivot(index='Year', columns='MonthNum', values='Passengers')
```

```
plt.figure(figsize=(12,6))
sns.heatmap(pivot, annot=True, fmt=".0f", cmap="YlGnBu")
plt.title('Monthly Airline Passengers Heatmap')
plt.xlabel('Month')
plt.ylabel('Year')
plt.show()
```



✓ 5. Interactive Time Series Plot

Description:

An interactive line plot for exploring the time series data.

Features used:

- Month (x-axis)
- Passengers (y-axis)

Observation:

- Interactive features allow zooming and hovering for precise values.

```
fig = px.line(df, x='Month', y='Passengers', title='Monthly International Airline  
fig.show()
```



Monthly International Airline Passengers (Interactive)

